VT2350

A VT100 terminal on an RP2350

User Manual, November 2026

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1 What it is

VT2350 turns an RP2350 board into a stand-alone serial terminal. You give it an HDMI monitor, a USB keyboard and a serial line, and it shows whatever is at the other end: a PicoMite, a Linux machine, a router's console or anything else that talks to a terminal. No PC is needed, and no PSRAM.

- 640 x 480 in 256 colours on HDMI, with seven font sizes from 128 x 60 characters down to 40 x 15.
- Two UARTs, or up to four USB serial devices, as the line.
- A USB keyboard in six layouts, and a USB mouse.
- VT100 and VT52, with xterm's 256 colours, UTF-8, CP437 and Latin-1, double-width and double-height lines, an alternate screen, mouse reporting and scrollback.
- One firmware for boards with different HDMI wiring.

2 What you need

- An RP2350 board with HDMI on GP12 to GP19, such as a Pico 2 with an HSTX DVI sock.
- An HDMI monitor or television.
- A USB keyboard. A mouse is optional.
- A USB hub, if you want more than one USB device: a keyboard with a mouse or a USB serial device needs one.
- Something to talk to, on a UART or a USB serial device.

The connections

Pin	What it is for
GP0	UART0 transmit: goes to the other end's receive
GP1	UART0 receive: comes from the other end's transmit
GP2	The bell, a tone for a piezo sounder or an amplifier (see section 11)
GP4	UART1 transmit
GP5	UART1 receive
GP6, GP7	Optional straps that choose the HDMI wiring (see section 10)
GP12 to GP19	HDMI
USB port	USB host, for the keyboard, mouse and USB serial devices

The UARTs are 3.3 V logic. Join the two grounds, and cross the lines over: transmit on one end goes to receive on the other. A real RS-232 port needs a level converter.

The USB port has to supply 5 V. It is working as a host, so the power for the keyboard and hub comes from your board. The next part shows how to do that on a Pico 2.

The smallest build

This is the least you can build a complete terminal from. There is nothing to wire but the 5 V supply.

Part	What it does
Pico 2, with headers	The terminal
HSTX DVI sock	The HDMI socket, plugged on to GP12 to GP19
Micro-USB to USB-A socket adapter	Turns the Pico 2's USB socket into a full-size one
USB hub, unpowered	Gives you sockets for a keyboard, a mouse and a USB serial device
5 V supply	Powers the Pico 2, and through it the hub and all that is on it

1. Install the firmware first, with the Pico 2 on its own (section 3).
2. Fit the sock and connect the monitor. The sock is wired the way the terminal expects, so it needs no straps.
3. Plug the adapter into the Pico 2, the hub into the adapter, and the keyboard into the hub.
4. Connect the 5 V supply: positive to **VBUS (pin 40)** and negative to **GND (pin 38)**.

The supply has to go to VBUS. VSYS (pin 39) will power the Pico 2, but the power doesn't reach the USB socket from there, and the keyboard stays dead. Use a supply rated at 2 A.

Never connect the 5 V supply and a PC at the same time. VBUS is joined straight to the USB socket, so the two supplies would be connected together. Disconnect the supply before you plug the Pico 2 into a PC to install or update the firmware.

3 Installing it

1. Hold down the board's BOOTSEL button and connect its USB port to a PC. A drive called RP2350 appears.
2. Copy `VT2350.uf2` to that drive. The board restarts by itself and the drive goes away.
3. Unplug the PC, and connect the keyboard or hub, the monitor and the serial line.

Updating it

Disconnect any 5 V supply, hold down BOOTSEL, and connect the board's USB port to a PC instead of the hub. Then copy the new UF2 to the RP2350 drive. On a board that stays powered while its USB port moves to a PC, **Firmware update** in the setup menu does the same without BOOTSEL.

Saved settings stay, unless the new version saves different things, when it starts from the defaults.

4 First use

Power it up. The screen clears, and the last line is the status line:

```
UART0 115200 8N1  |  80x33  |  Shift-F12 setup
```

That is the line in use, its speed and format, and the size of the terminal. As it comes, the terminal uses UART0 at 115200, 8 bits, no parity, one stop bit.

If the other end is already running, press Enter and you should see it answer. If it needs a different

speed or port, press **Shift-F12** to open the setup menu.

If there is no picture, or the colours are wrong, your board's HDMI wiring differs from the default: go to section 10.

5 The setup menu

Shift-F12 or **Pause** opens the menu.

Key	In the menu
Up, Down	Move to another line
Left, Right	Change the value
Enter	Change the value, or do what the line says
Esc, Shift-F12, Pause	Leave the menu

Changes take effect when you leave the menu, and last until the power goes off. To keep them, choose **Save settings**. The HDMI wiring is the exception: it takes effect at once, and has its own way of being kept (section 10).

Line	Choices	As it comes
Port	UART0 (GP0, GP1), UART1 (GP4, GP5), USB1 to USB4	UART0
Speed	300, 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600	115200
Format	8N1, 8E1, 8O1, 7E1, 7O1, 8N2	8N1
Flow control	None, XON/XOFF (section 9)	None
Font	5x8, 6x12, 8x12, 8x14, 8x16, 12x24, 16x32	8x14
Columns	All, 40, 80, 100, 120	All
Rows	All, 24, 25, 30, 36, 40, 43, 48, 50, 60	All
Status line	On, Off	On
Characters	CP437, Latin-1, UTF-8: how the terminal reads bytes above 127	CP437
Text colour	White, Green, Amber	White
Cursor	Block, Underline, Bar	Underline
Keyboard	US, UK, German, French, Spanish, Belgian	US
Backspace sends	BS (8), DEL (127)	BS (8)
Enter sends	CR, CR LF	CR
LF received as	LF, CR LF	LF
Local echo	Off, On: On shows what you type, for a host that does not	Off
Bell	Tone, Flash, Off	Tone
HDMI wiring	Section 10	2,0,6,4

Line	What it does
Reset the terminal	Clears the screen and the scrollback and resets the terminal, then leaves the menu
Save settings	Keeps everything in the menu through a power cycle
Restart	Restarts the board
Firmware update	Goes to the bootloader, for a new UF2

Fonts and screen sizes

The font sets how much fits on the screen. The status line takes one row. Columns and Rows make the terminal smaller than that, if a program needs a set size: asking for more than the font allows gives you the most it can.

Font	With the status line	Without it
5x8	128 x 59	128 x 60
6x12	106 x 39	106 x 40
8x12	80 x 39	80 x 40
8x14	80 x 33	80 x 34
8x16	80 x 29	80 x 30
12x24	53 x 19	53 x 20
16x32	40 x 14	40 x 15

6 The keyboard

Almost every key goes to the host, including all twelve function keys and Ctrl-C. Only Shift-F12, Pause and the scrollback keys stay with the terminal.

Key	What it does
Shift-F12, Pause	Opens the setup menu
Shift-Page Up, Page Down	Moves back and forward through the scrollback
Shift-Home, Shift-End	Goes to the oldest line kept, and back to the live screen
F1 to F12	The usual function key codes
Shift-F3 to Shift-F10	VT220's F13 to F20, which a PicoMite reads as its own Shift-F keys
Home, End	ESC [1 ~ and ESC [4 ~, which are what a PicoMite expects
Alt with a key	ESC, then the key
Backspace	As the Backspace sends line in the menu
Enter	As the Enter sends line in the menu

The arrows, Insert, Delete, Page Up, Page Down and Shift-Tab send the usual codes. In VT52 mode the arrows are `ESC A` to `ESC D`, and F1 to F4 are PF1 to PF4.

7 Scrollback and the mouse

Lines that scroll off the top are kept for as long as there is memory for them. How many depends on the size of the terminal, as a wider screen uses more for each line: about 160 lines at 80 x 33, and about 48 at 128 x 59.

Use Shift with Page Up, Page Down, Home and End, or the mouse wheel, which moves three lines at a click. Any other key takes you back to the live screen.

That is all the mouse does until a program asks for it. When a host program turns on mouse reporting, a pointer appears, and movement, the buttons and the wheel go to the program. The wheel stops scrolling the terminal until the program turns reporting off.

8 Using it with a host

With a PicoMite

Join the terminal to the PicoMite's serial console, or plug a PicoMite into the hub as a USB serial device and choose USB1. The defaults suit it: CP437 characters, and 115200 on a UART. The keys are sent the way the PicoMite expects them, for EDIT and FM.

The PicoMite tells the terminal what size it wants at each EDIT and FM, and at OPTION DISPLAY. If that does not fit, the terminal moves to a smaller font until it does, and gives up the status line if the host needs every row.

With anything else

- For Linux and other Unix hosts, set **Characters** to UTF-8. `TERM=vt100` is the safe choice, and `xterm-256color` is worth trying for colour.
- If each line starts where the last one ended, set **LF received as** to CR LF.
- If the host needs a line feed after Enter, set **Enter sends** to CR LF.
- If Backspace does the wrong thing, change **Backspace sends**.

There is not room for 132 columns. A program that asks for them gets as many as the font allows.

9 The serial lines

USB serial devices

Up to four can be connected, as USB1 to USB4 in the order they are found. Devices that follow the USB CDC standard work, including PicoMites, as do those with FTDI, CP210x, CH34x and PL2303 chips.

If the device you choose is not there, the status line says `no USB serial device there`, and the terminal keeps looking for it. If it is unplugged, the status line says so, and the port opens again when it comes back.

How fast it keeps up

What takes the time is lines, not bytes, as each new line scrolls the screen. Short lines sent without a pause are the hardest thing a host can do. With 64-byte lines:

Font	460800	921600
8x12, 8x14, 8x16	Nothing lost	Nothing lost
6x12	Nothing lost	Loses data: it keeps up with about 73 KB/s
5x8	Nothing lost	Loses data: it keeps up with about 65 KB/s

The two largest fonts have not been measured. The terminal holds 16 KB of incoming data, so short bursts are safe in any case, and longer lines are easier. A USB serial device has been measured at 46 to 81 KB/s with nothing lost.

To be sure of losing nothing, set Flow control to XON/XOFF. The terminal then asks the host to pause when its buffer fills. In the hardest case, 921600 in the 5x8 font, this lost nothing, at about 67 KB/s. The host has to support XON/XOFF too.

10 HDMI wiring

Boards put the four HDMI pairs on GP12 to GP19 in different orders. The wiring is four numbers, the same as the PicoMite's `OPTION HDMI PINS`, for the clock, D0, D1 and D2 pairs. Each number is 0 for GP12 up to 7 for GP19, and gives the pair's positive pin. There are 384 possible wirings, and any of them can be used.

The terminal takes the wiring from the first of these that gives one: the straps, then the saved wiring, then the default, 2,0,6,4.

Straps

Tie GP6 and GP7 to ground or 3.3 V, or leave them open. They are read each time the board starts.

GP6	GP7	Wiring	Board
Ground	Ground	2,0,6,4	The HSTX sock, and the PicoMite's default
Ground	Open	1,3,5,7	Pico Computer 3
Open	Ground	1,3,7,5	Olimex
3.3 V	Open	3,1,7,5	2,0,6,4 with every pair the other way round
Open	3.3 V	0,2,4,6	1,3,5,7 with every pair the other way round
3.3 V	3.3 V	0,2,6,4	1,3,7,5 with every pair the other way round

Open on both, or ground on one and 3.3 V on the other, chooses nothing. A wiring set by the straps can't be changed in the menu.

With no picture

The keyboard is enough. You can do this without being able to see anything:

1. Press **Shift-F12** to open the menu.

2. Press **W**. The next wiring in the list above takes effect at once.
3. Keep pressing **W** until the picture is right.
4. Press **Enter** to keep that wiring. It is saved straight away. Press any key to clear the message.

A new wiring goes back by itself after 10 seconds unless you keep it, and Esc takes it back at once, so a wrong one can't lose the picture for good.

A picture is not enough to go on. The wrong wiring can give one. Look at the message: it shows the words Red, Green and Blue in those colours, on a light grey box.

What you see	What is wrong
No picture	The clock or D0 is on the wrong pair
Red and green have changed places	D1 and D2 are swapped
Colours inverted, like a negative	The pairs are the wrong way round. Other monitors may show no picture for this

A wiring that is not in the list

With a picture, go to the **HDMI wiring** line in the menu. Left and Right go through the list. Enter asks for four numbers, each 0 to 7, and each of the four needs a pair of its own. Then Enter keeps the wiring, as above.

11 The bell

Tone gives a 790 Hz beep lasting 150 ms on GP2. It is a logic-level square wave: use a piezo sounder, or an amplifier for a speaker. **Flash** inverts the screen's colours for a moment instead. **Off** ignores the bell.

12 If something is wrong

What happens	What to try
No picture, or the wrong colours	The HDMI wiring: section 10
The keyboard does nothing	Check the USB port is getting 5 V: on a Pico 2 the supply must go to VBUS, not VSYS. Give the board about 12 seconds after it starts to find what is on a hub. If it still does nothing, try a powered hub
Nothing appears from the host	Check the status line shows the right port and speed. On a UART, check transmit and receive are crossed over and the grounds are joined
Rubbish on the screen	The Speed or Format differs from the host's
Every line starts where the last one ended	Set LF received as to CR LF
Lines are double spaced	Set LF received as to LF
Boxes and lines show the wrong characters	Set Characters to what the host sends: CP437 for a PicoMite, UTF-8 for Linux
You see each character twice	Set Local echo to Off
You can't see what you type	The host is not echoing: set Local echo to On

What happens	What to try
Text goes missing in a fast listing	Set Flow control to XON/XOFF, or use a larger font
The font changed by itself	The host asked for a size that needs it: section 8
The mouse wheel stopped scrolling	A program has asked for the mouse. It stops when the program turns reporting off
Settings are lost at power-off	Choose Save settings in the menu
USB1 shows no USB serial device	Check it is plugged in. If several are, it may be USB2 to USB4

13 Licence and credits

VT2350 is free software under the MIT License. The source is at github.com/UKTailwind/VT2350.

It uses libvterm for the terminal, the Spleen fonts, TinyUSB and the Raspberry Pi Pico SDK. Its HDMI, keyboard and mouse code comes from the PicoMite, and the terminal from the Colour Maximite 2's VT100 application. The file `PROVENANCE.md` in the source gives the details and the licences.